# Working with data: inspect, fix, create

## Programming & Data Management, Part II (Python) · Session 2

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# Today's goals

By the end of today you can:

1. Apply the intake checklist to a variable: type, missing values, range
2. Fix types and missing values, including sentinel codes such as 999
3. Create, subset, sort and aggregate variables, and summarise data in a table and a figure

# From last time

Answer on your own first, then compare with a neighbour

1. You have cleaned a dataset. Where does your code save it, and why not in `data/raw/`?
2. Write the pandas line that gives mean mpg by origin.

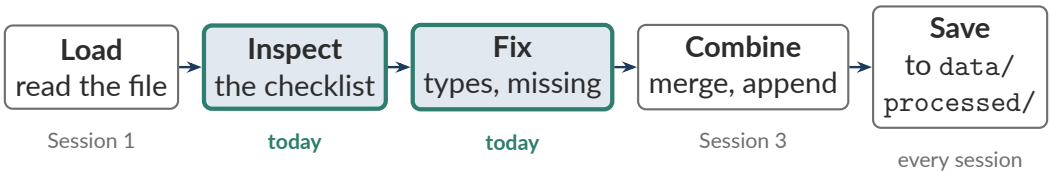
# Before we start: open the course folder, not the lab

`na446a-starter/` ← **open this folder in VS Code**
- `.venv/` the course Python, with pandas
- `s1-lab/`
- `s2-lab/` ← today's lab goes *inside*

- VS Code asks for *ipykernel*? Not *Install*: *Change Kernel*, refresh the list (↻), then `na446a … .venv`.

# Every data task follows one recipe

Today: inspect and fix, then create and aggregate




- Today's two errors never crash a script: a code read as a number, and a missing value counted as a zero.

# Six terms for today

| | |
|---|---|
| `dtype` | the type of a column: `float64` (number), `int64`, `str` (text) |
| `NaN`, `<NA>` | pandas' missing value; Stata's `.` |
| sentinel code | a number that means "missing", such as $-9$ or 999 |
| True/False mask | a True/False column that picks rows: `cars["mpg"] > 40` |
| `groupby(...).agg` | one row per group: Stata's `collapse` |
| `groupby(...).transform` | a group value on every row: Stata's `egen ..., by()` |

- A sentinel code looks like data. Only the range check, or the codebook, gives it away.

# The intake checklist: ten questions for every variable

Today we practise the four in green

1. Name, label and unit
2. Type: number, text or date?
3. Missing values, incl. sentinel codes
4. Range: possible min and max?
5. Distribution and outliers
6. Categories: levels and typos
7. Dates: read as dates, no gaps
8. IDs: unique, never blank
9. Logic across variables
10. Documentation of every change

- Ask all ten before any regression. Items 7–8 return in Session 3.

▷ The checklist in code

# Steps 1–6: check, find, fix

| | Check | We find | Fix |
|---|---|---|---|
| horsepower | Step 1: types | text (“?”) | Steps 2–3: `to_numeric` |
| mpg | Steps 4–5: missing, range | 999 codes | Step 6: `mask` |

- Two problems, one rhythm: check, find, fix, then count again.

# Step 1, check: are the types right?

```
RAW = Path("data/raw")
cars = pd.read_csv(RAW / "auto_messy.csv")
cars.dtypes.head(5)
```

```
mpg             float64
cylinders         int64
displacement    float64
horsepower          str
weight            int64
dtype: object
```

- Horsepower is a number, yet pandas read it as text (`str`): at least one value in the column is not a number.

`auto_messy.csv`: the Session 1 file with two typical problems planted. Stata: `describe`.

# Step 2, find: which values are not numbers?

```
# trial conversion into hp; cars is unchanged
hp = pd.to_numeric(cars["horsepower"], errors="coerce")
# isna = is missing: the rows that failed
cars.loc[hp.isna(), ["name", "horsepower"]]
```

| | name | horsepower |
|---|---|---|
| 56 | toyota corona hardtop | ? |
| 71 | amc matador (sw) | ? |
| 114 | chevrolet monte carlo s | ? |
| 151 | chevrolet nova | ? |

`coerce`: unreadable values become `NaN`. `.loc[rows, columns]`.

- Four cars have “?” for unknown horsepower. Look before you convert.

# Step 3, fix: horsepower becomes a number

```
cars["horsepower"] = hp
print(cars["horsepower"].dtype)
print(cars["horsepower"].isna().sum())
```

```
float64
4
```

- Horsepower is now a number, and the 4 “?” are real missing values. Count before and after: `coerce` turns anything it cannot read into missing, silently.

Stata: `destring horsepower, replace force` (“4 missing values generated”).

# Step 4, check: how many values are missing?

```
# after sum(), each row is a variable; head(4): first four
cars.isna().sum().head(4)
```

```
mpg             0
cylinders       0
displacement    0
horsepower      4
dtype: int64
```

`isna()` is True where a value is missing; `sum()` counts the Trues (True = 1) in each column. It only sees what pandas already knows is missing.

- According to this, mpg has no missing values. Is that true? The next check decides.

Stata: `misstable summarize`, or `count if missing(mpg)`.

# Step 5, check: is the range possible?

```
stats = cars[["mpg", "horsepower"]].describe()
stats.loc[["count", "mean", "min", "max"]].round(1)
```

```
          mpg  horsepower
count   392.0       388.0
mean     30.9       104.3
min       9.0        46.0
max     999.0       230.0
```

**Ford Pinto (sw), 1972, row 79** mpg 999: one of 3 cars where "missing" was written as a number.

– 999 is a sentinel code, and it pulls the mean up from 23.4 to 30.9.

# Step 6, fix: the 999 code becomes missing

```
is_code = cars["mpg"] == 999
print(is_code.sum())
# mask = tape over the 999s: they become missing, rows stay
cars["mpg"] = cars["mpg"].mask(is_code)
print(cars["mpg"].isna().sum())
print(cars["mpg"].mean().round(1))
```

```
3
3
23.4
```

- All 392 cars kept, 3 now missing; the mean skips them: 23.4.

Stata: `mvdecode mpg, mv(999)` (“3 missing values generated”).

# A car with missing mpg: what does `mpg > 25` give it?

Your bet first: hands up. The answer is on the next slide.

A. 1: a high-mpg car
B. 0: a low-mpg car
C. Missing

# Stata says 1, pandas says 0, and both are wrong

**Stata:** `gen high = mpg > 25`

Missing counts as larger than any number, so the car gets **1**: 157 high-mpg cars

**pandas:** `(cars["mpg"] > 25).astype(int)`

`NaN > 25` is False, so the car gets **0**: 154 high-mpg cars, and 3 unknowns counted as low

> **Ford Pinto (sw)** Stata calls it a high-mpg car, pandas a low-mpg car. We do not know its mpg.

- Keep missing values missing: Step 8 shows how. Verified in both languages.

# Steps 7–13: the question each step answers

| | Question | Tool (Stata) |
|---|---|---|
| 7 | mpg in km per litre, origin as names? | new column, `map` (`gen`, `label`) |
| 8 | high mpg yes/no, Pinto still unknown? | `where` (`gen ... if`) |
| 9 | which cars do more than 40 mpg? | `.loc`, `sort_values` (`keep`, `gsort`) |
| 10 | how do the regions compare? | `groupby().agg` (`collapse`) |
| 11 | how does each car compare with its region? | `transform` (`egen, by()`) |
| 12–13 | how do we show it? | a figure, saved by code |

- Every step answers one question. Say the question before you write the code.

# Step 7: mpg in km per litre, origin as names

```
cars["kpl"] = cars["mpg"] * 0.425144
names = {1: "US", 2: "Europe", 3: "Japan"}
cars["region"] = cars["origin"].map(names)
cars[["mpg", "kpl", "region"]].head(3).round(1)
```

```
    mpg  kpl region
0  18.0  7.7     US
1  15.0  6.4     US
2  18.0  7.7     US
```

- A new column is computed for every row at once, with no loop. `map` turns codes into labels: check them against the codebook (Lab 1's bug).

Stata: `gen kpl = mpg * 0.425144`; `label define` and `label values`. One mile per US gallon is 0.425144 km per litre.

# Step 8: a high-mpg dummy, unknowns stay unknown

```
ok = cars["mpg"].notna()
high = (cars["mpg"] > 25).astype("Int8")
cars["high_mpg"] = high.where(ok)
cars["high_mpg"].value_counts(dropna=False)
```

```
high_mpg
0       235
1       154
<NA>      3
Name: count, dtype: Int64
```

- 235 + 154 + 3 = 392: every car counted, and the Pinto stays unknown.

`where(ok)` keeps a value where `ok` is True, missing elsewhere (the mirror of `mask`); `Int8` is an integer type that can hold `<NA>`. Stata: `gen high = mpg > 25 if !missing(mpg)`.

# Step 9: which cars do more than 40 mpg?

```
fast = cars.loc[cars["mpg"] > 40,
                ["name", "mpg"]]
fast = fast.sort_values("mpg",
                        ascending=False)
fast.head(3)
```

| | name | mpg |
|---|---|---|
| 320 | mazda glc | 46.6 |
| 327 | honda civic 1500 gl | 44.6 |
| 323 | vw rabbit c (diesel) | 44.3 |

- A mask picks the rows (8 cars), a list the columns; `cars` is unchanged.

Stata: `gsort -mpg`, `list name mpg if mpg > 40`. pandas returns a sorted copy (assign it); row labels travel with the rows; missing values go last.

# Step 10: how do the regions compare? (agg)

```
by_region = cars.groupby("region").agg(
    n=("mpg", "count"),
    mean_mpg=("mpg", "mean"),
    mean_hp=("horsepower", "mean"))
by_region.round(1)
```

| region | n | mean_mpg | mean_hp |
|---|---|---|---|
| Europe | 67 | 27.6 | 80.6 |
| Japan | 79 | 30.5 | 79.6 |
| US | 243 | 20.0 | 118.9 |

- One row per region, as Stata's `collapse`; the means of Session 1.

▷ count or size?

# Step 11: each car against its region (`transform`)

```
grp = cars.groupby("region")["mpg"]
cars["region_mpg"] = grp.transform("mean")
cars["gap"] = cars["mpg"] - cars["region_mpg"]
show = cars[["mpg", "region_mpg", "gap"]]
show.head(3).round(1)
```

```
    mpg  region_mpg  gap
0  18.0        20.0 -2.0
1  15.0        20.0 -5.0
2  18.0        20.0 -2.0
```

- Still 392 rows: each car gets its region's mean, so you can compare the car with its group.

Stata: `bysort origin: egen region_mpg = mean(mpg)`.

# Check your understanding

Answer on your own first, then compare with a neighbour

You want each car's horsepower relative to the mean horsepower of cars with the same number of cylinders. `agg` or `transform`? How many rows does the result have?

# Step 12: show the region table as a figure

```
m = by_region["mean_mpg"].sort_values()
fig, ax = plt.subplots(figsize=(6, 2.4))
ax.barh(m.index, m, color="0.35")
ax.set_xlabel("Miles per US gallon (mean)")
ax.spines[["top", "right"]].set_visible(False)
```

`fig` is the page, `ax` the drawing area: every change is `ax.`something. Grey bars, sorted, read as a ranking and survive black-and-white printing.

- Draw from the table you already checked, never from numbers typed by hand.

Needs `import matplotlib.pyplot as plt` at the top of the script. Stata: `graph hbar (mean) mpg, over(origin)`.

# Step 13: a title that states the finding, then save

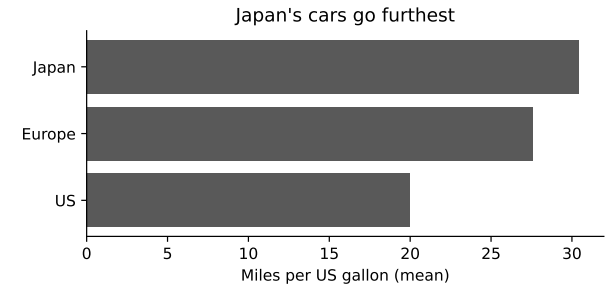
```
ax.set_title("Japan's cars go furthest")
fig.text(0, -0.15, "Source: auto.csv, 389 cars")
for ext in ["png", "pdf"]:
    fig.savefig(f"output/figures/mpg.{ext}",
                bbox_inches="tight")
```

The title states the finding, not the variable; the source line names the data and sample. The loop saves the figure twice: PDF for papers, PNG for slides.

- Figures go to `output/figures/`, written by code, so a rerun redraws them.

Stata: `graph export "output/figures/mpg.pdf", replace`.

# What makes a figure publication-ready




Source: auto.csv, 389 cars

The figure exactly as Step 13 saved it.

1. The title states the finding, not the variable
2. Axes labelled, with units
3. A source line: data and sample
4. Grey that survives black-and-white printing
5. Nothing else: no box, grid, 3D or legend for one series

- If a reader must read the text to understand the figure, the figure is not done.

# Today's lab

1. Fill in the TODOs in `s2_lab.py`, one cell at a time *goals 1–3*
2. Check your steps: run `check_s2.py`, fix what it flags *goals 1–3*
3. Ask an AI, then check it: paste your *intake output* (not the data) into a chatbot and ask what problems it shows. Test two of its claims with one line of code each; note claim, test, true or false *rules 3–4*

– A chatbot can read an output well and still be wrong about the data behind it. Your line of code decides.

Never paste personal or register data into a chatbot. Stuck? Use the course tutor prompt (course site, AI page): it gives hints, not answers.

# Summary

1. The checklist first: `dtypes`, `isna().sum()`, `describe()` catch a text column and a sentinel code
2. Fix with `to_numeric(errors="coerce")` and `mask`; count missing values before and after every fix
3. Create with `=`, `map` and `where`; pick rows with `.loc`; `agg` gives one row per group, `transform` one value per row; a figure states its finding and its source

**Before next time:** the Titanic intake exercise (course site), with its self-check; about 90 minutes. Solution after three days.

# Test yourself before next time

Without AI: this is what you should manage on your own

1. `income` was read as `str`. Name two likely causes and the line that fixes it.
2. One value was `"1,200"`. What does `pd.to_numeric(..., errors="coerce")` give it, and how would you notice?
3. Write the line that turns the codes $-9$ and 999 in `age` into missing values.
4. Do `cars["mpg"] <= 25` and `~(cars["mpg"] > 25)` pick the same cars? Why not?
5. Name three things that make a figure publication-ready.

# Appendix

# The checklist in code

| Question | pandas | Stata |
|---|---|---|
| 1 Name, label, unit | `df.columns`, a codebook file | `label variable` |
| 2 Type | `df.dtypes`, `pd.to_numeric` | `describe`, `destring` |
| 3 Missing, sentinels | `df.isna().sum()`, `mask` | `misstable sum`, `mvdecode` |
| 4 Range | `df.describe()` | `summarize` |
| 5 Distribution, outliers | `df["x"].hist()`, `quantile` | `histogram`, `centile` |
| 6 Categories | `value_counts(dropna=False)` | `tab x, missing` |
| 7 Dates | `pd.to_datetime` | `date()`, `tsset` |
| 8 IDs and keys | `df["id"].is_unique` | `isid id` |
| 9 Cross-variable logic | `assert (df["wage"] > 0).all()` | `assert wage > 0` |
| 10 Documentation | comments, README, codebook | `notes`, do-file |

# `count` or `size`? And two kinds of missing

- `count` counts non-missing values: Europe has 67 cars with mpg
- `size` counts rows, missing or not: Europe has 68 cars
- Report the one you mean; a table of means should show the `count`

- `NaN` is the missing value of ordinary number columns (`float64`): `NaN > 25` is False
- `<NA>` is the missing value of types with a capital letter (`Int8`, `Float64`): `<NA> > 25` is `<NA>`, so missing stays missing

# One more trap: “25 or less” is not “not above 25”

- `len(cars[cars["mpg"] <= 25])` gives 235
- `len(cars[~(cars["mpg"] > 25)])` gives 238. The tilde (~) means *not*: it flips True and False (Stata’s `!`)
- The difference is the 3 cars with missing mpg: `NaN > 25` is False, and *not* False is True, so they are kept
- In Stata both give 235, since missing counts as greater than 25

- Write the condition you mean, and count the rows it keeps.